MISSION YUVA

Loan Analytics & Financial Inclusion Report

Government of Jammu & Kashmir — Youth Entrepreneurship Initiative

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Sector	Banking / MSME Finance / Government Scheme Analytics

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2. Executive Summary

Mission Yuva is a Government of Jammu & Kashmir initiative providing collateral-free loans to youth entrepreneurs across key economic sectors. This report presents a data-driven analysis of 2,610 loan applications aggregated across five districts — Doda, Pulwama, Reasi, Samba, and Shopian — amounting to a combined portfolio of Rs. 546 Crore. Four Power BI dashboards dissect loan distribution, repayment health, applicant inclusion, and financial risk metrics.

2,610	Rs. 546 Cr	51 / 100	188 Days
Total Applications	Total Loan Disbursed	Avg Risk Score	Avg TAT

Problem

The lending authority lacks real-time visibility into which districts, sectors, and demographic segments carry elevated repayment risk, limiting its ability to intervene proactively and allocate resources equitably.

Approach

Raw application data was cleaned, KPIs were engineered, and four interconnected dashboards were built covering (a) loan overview, (b) repayment & risk, (c) applicant profiling, and (d) financial deep dive. Descriptive statistical segmentation and district-level TAT analysis were performed to surface actionable insights.

Key Highlights

- 33% of applications (863) resulted in late payments; 399 ended in default — a combined stress rate of 47.8%.
- Tourism and Agriculture sectors carry the highest default rates (>15%), warranting priority monitoring.
- Average EMI-to-income ratio stands at 0.30, indicating moderate but manageable borrower leverage.
- Doda leads in absolute loan volume; the DLIC approval stage is the single biggest TAT bottleneck.
- 1,501 specially abled applicants represent a significant inclusion segment requiring tailored support.

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3. Sector and Business Context

Sector Overview

The MSME / youth entrepreneurship lending sector in J&K operates within a challenging terrain — geographically dispersed borrowers, limited formal credit histories, and sector volatility (especially in Tourism and Agriculture). Mission Yuva targets eight sectors: Agriculture, Healthcare, IT, Manufacturing, Retail, Services, Tourism, and Others.

Decision-Maker / Stakeholder

- District-level SLBC (State-Level Bankers' Committee) officers — approvals and TAT management.
- J&K Mission Yuva Nodal Agency — scheme policy and borrower eligibility.
- Partnering banks (SBDU, DLIC, Sanctioning bodies) — credit risk and disbursement.
- Beneficiaries — youth entrepreneurs aged 18–35 across rural and urban localities.

Why This Problem Matters

High default rates erode the scheme's revolving fund, reduce future cohort coverage, and signal potential structural mismatches between borrower capacity and loan sizing. Early identification of at-risk segments enables targeted credit counselling, restructuring, and field-level intervention before default crystallises.

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4. Problem Statement and Objectives

Formal Problem Definition

Given the Mission Yuva loan application dataset, identify the key drivers of late payment and default, quantify the TAT bottlenecks across the five-stage disbursement pipeline, and characterise the applicant demographic to inform equitable loan allocation and risk mitigation strategies.

Scope

- Geography: 5 districts of J&K — Doda, Pulwama, Reasi, Samba, Shopian.
- Time: Historical cohort of 2,610 loan applications.
- Sectors: 8 economic sectors as captured in application data.
- Out of scope: Macro-economic forecasting; individual borrower credit scoring model.

Success Criteria

- Dashboard suite covering all 11 defined KPIs, filterable by district, gender, sector, and repayment status.
- Identification of at least 5 high-confidence, decision-ready insights.
- Visualisation of complete TAT pipeline across all 5 process stages per district.

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5. Data Description

The dataset was sourced from the Mission Yuva scheme administrative database, containing one row per loan application with the following key attribute groups:

Attribute Group	Key Columns	Notes
Applicant Identity	Application ID, Name, Gender, Date of Birth	Gender: 4 categories incl. Unknown
Geography	District Name, Residential Type (Rural/Urban)	5 districts
Loan Financials	Loan Amount INR, Project Cost INR, Monthly Income	Used for EMI Ratio calc
Sector & Enterprise	Sector, Enterprise Type	8 sectors; 4 enterprise types
Risk & Credit	Risk Score (0-100), Credit Score	Avg Risk=51, Avg Credit=50
Repayment	Repayment Status, Late Payment Flag	On-Time / Late / Default / Unknown
TAT Pipeline	Days: BHD Verify, SBDU Verify, DLIC Approve, Bank Sanction, Disburse	5-stage pipeline
Inclusion	Specially Abled Type	Hearing/Locomotor/Multiple/Visual/Unknown

Dataset Size: 2,610 application records.

Data Quality Issues Observed

- Gender field contains 'Unknown' category — treated as a separate segment, not imputed.
- Residential Type has 'Unknown' records — retained for completeness.
- Specially Abled Type has 1,109 'Unknown' entries — limits disability-disaggregated insights.
- Risk Score and Credit Score both average ~50/51, suggesting a normalised scoring method.

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6. Cleaning and Transformation

Major Cleaning Steps

- Standardised sector names (capitalisation and whitespace trimming).
- Converted currency fields from raw INR integers to Cr / K notations for display.
- Derived Repayment Status binary flags (Late=1/0, Default=1/0) from the status field.
- Computed TAT per stage as numeric days; records with negative TAT values were flagged.
- Created Risk Score bins (0-10, 10-20, ..., 90-100) for histogram visualisation.
- Calculated EMI Ratio = Loan Amount / (Applicant Monthly Income x 60) as affordability metric.

Assumptions Made

- Total Loan (Rs. 546 Cr) represents committed/disbursed amounts, not project cost.
- EMI Ratio denominator assumes a 60-month (5-year) repayment horizon.
- Default Rate by Sector = Count of Default applications / Total applications in that sector.
- TAT days are calendar days, not business days.

Output Dataset

Cleaned flat file retaining all 2,610 rows; 6 derived columns added; loaded into the Power BI data model.

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7. KPI Framework

The following 11 KPIs were defined, engineered, and visualised across the four dashboards:

KPI	Name	Formula	Value	Rationale
KPI-1	Total Applications	COUNT(Application ID)	2,610	Overall scheme reach
KPI-2	Total Loan Disbursed	SUM(Loan Amount INR)	Rs.546 Cr	Portfolio size
KPI-3	Avg Risk Score	AVG(Risk Score)	51	Portfolio-level risk
KPI-4	Avg TAT	AVG(Sum of 5 TAT stages per application)	188 days	End-to-end process speed
KPI-5	Late Payments Count	COUNT WHERE Repayment Status = 'Late'	863	Early repayment stress signal
KPI-6	Avg Credit Score	AVG(Credit Score)	50	Borrower creditworthiness
KPI-7	Total Defaulted Loans	SUM(Loan) WHERE Status = 'Default'	Rs. 82 Cr	Credit loss exposure
KPI-8	Female Applicants	COUNT WHERE Gender = 'Female'	2,610*	Gender inclusion metric
KPI-9	Specially Abled Count	COUNT WHERE Specially Abled = 'Yes'	1,501	Disability inclusion metric
KPI-10	Rural Count	COUNT WHERE Residential = 'Rural'	2,610*	Rural outreach measure
KPI-11	Avg EMI Ratio	AVG(Loan Amount / (Income x 60))	0.30	Borrower affordability check

** KPI-8 and KPI-10 display 2,610 at the 'All' filter level. Segment-specific counts appear after applying gender/residential filters.*

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8. Exploratory Analysis

8.1 Loan Overview — Dashboard 1

Dashboard 1 provides the executive view of overall portfolio health.

2,610	Rs. 546 Cr	51	188 Days
Total Applications	Total Loan	Avg Risk Score	Avg TAT

- **Sector x Gender Heatmap:** Male applicants dominate most sectors. Healthcare and Services show relatively higher female participation. Agriculture has the widest absolute volumes across all gender groups.
- **Risk Score Distribution:** Scores peak sharply in the 50-60 bin (~350 applications), with a roughly symmetric spread, suggesting a normalised scoring model. Very few applicants score below 20 or above 90.
- **TAT Analysis:** Pulwama consistently shows the longest individual stage durations. 'Days to DLIC Approval' is the longest stage across most districts, pointing to a clear bottleneck in district-level government approval.
- **District Loan Overview:** Doda leads with the highest total loan amounts across multiple tranches. Shopian records the smallest aggregate but comparable application count, suggesting smaller per-loan ticket sizes.

8.2 Repayment & Risk Analysis — Dashboard 2

863	50	Rs. 82 Cr	399
Late Payments	Avg Credit Score	Defaulted Loans	Default Count

- **Repayment Status Split:** 889 On-Time (34%) | 863 Late (33%) | 399 Default (15%) | 459 Unknown (18%). Only one-third of the portfolio is performing cleanly.
- **Default Rate by Sector:** Tourism (~16%) and Agriculture (~15%) lead; Retail and Services are comparatively lower (~10%). IT sits mid-range (~13%).
- **Risk Score by Gender & District:** Female applicants in Reasi carry the lowest average risk score (49.1) while 'Other' gender in Shopian records 56.7 — the highest. Variation is modest (~7 points overall).

8.3 Applicant Profile & Inclusion — Dashboard 3

2,610*	1,501	2,610*
Female Applicants	Specially Abled	Rural Count

- **Gender x Residential:** Rural applicants outnumber urban across all gender groups. Female rural participation is notably high, underscoring the scheme's reach into non-urban households.
- **Enterprise Type by District:** Nano Entrepreneurs form the largest single enterprise type across all districts. MSME Sunrise Sector is the smallest enterprise category.

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- **Specially Abled Breakdown:** Out of 1,501 identified: Visual (395), Hearing (390), Multiple (366), Locomotor (350). A large 'Unknown' group (1,109) limits disability-specific policy targeting.

8.4 Financial Deep-Dive — Dashboard 4

0.30
Avg EMI Ratio

- **Loan Distribution:** The histogram peaks around Rs. 15 Lakh, with a rapid drop-off beyond Rs. 25 Lakh. Concentration in the Rs. 10-20 Lakh range aligns with the scheme's MSME ticket-size mandate.
- **Risk Score by Sector (Box Plots):** Median risk scores are consistent (~50) across all sectors; Agriculture and Tourism show slightly wider IQRs, reflecting higher within-sector borrower heterogeneity.
- **Loan vs Project Cost Scatter:** Strong positive linear relationship (R approximately 1). Majority of loans appear to be near-full funding of project costs, indicating minimal borrower equity contribution.
- **Avg Loan by District:** Doda leads average loan amount (~Rs. 22-23 Lakh), while Pulwama is at the lower end (~Rs. 18 Lakh), despite similar application volumes.

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9. Statistical Analysis

Method: Descriptive Segmentation + Cross-Tabulation

Given the analytical objective (portfolio monitoring, not prediction), the primary methods employed were descriptive statistics, cross-tabulation heat maps, and distribution analysis via histograms and box plots.

Analysis	Result	Business Interpretation
Default Rate by Sector	Tourism 16.3% | Agri 15.1% | IT 13.2% | Retail 10.4%	Tourism/Agri are highest-risk; Retail comparatively safer.
EMI Ratio Distribution	Mean = 0.30; range approx 0.20-0.45	Moderate leverage; borrowers below 0.40 considered sustainable.
TAT Stage Analysis	DLIC Approval is the longest stage across all districts	Government approval is the primary disbursement bottleneck.
Risk Score Spread	Near-normal distribution; mean = 51, approx SD = 20	Balanced risk portfolio; no extreme high-risk concentration.
Repayment Stress Rate	(863 Late + 399 Default) / 2,610 = 47.8%	Nearly half the portfolio is in some form of repayment stress.
Loan-Project Cost Correlation	Near-perfect positive linear (R approx 0.99)	Scheme funds almost entire project cost — minimal borrower equity.

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10. Dashboard Walkthrough

Dashboard 1 — Loan Overview

Provides the executive view of overall portfolio health. Four KPI cards give instant pulse on applications, loan value, average risk, and process speed. The Sector Breakdown heatmap and Risk Score histogram support strategic allocation decisions. TAT Analysis and District Overview serve district managers monitoring pipeline efficiency.

Filters: District Name, Gender, Repayment Status.

Dashboard 2 — Repayment & Risk Analysis

The credit risk monitoring view. KPIs spotlight late payments, credit quality, and defaulted loan value. Bar charts rank districts and sectors by repayment behaviour. The pie chart gives instant portfolio health proportion. The risk score cross-tab by gender and district helps identify intersectional risk pockets.

Filters: Gender, District, Repayment Status, Sector.

Dashboard 3 — Applicant Profile & Inclusion

The equity and inclusion lens. KPIs track female, specially-abled, and rural applicant counts. Stacked bar charts reveal the gender x residential mix; the enterprise-type distribution shows scheme usage by business category. The disability breakdown pie and sector x enterprise matrix inform policy targeting.

Filters: Gender, Residential Type, Enterprise Type, Sector.

Dashboard 4 — Financial Deep-Dive

The operational lending view. EMI Ratio KPI flags affordability. Loan distribution histogram and loan vs project cost scatter reveal ticket-size patterns. Sector-level risk box plots and district-level average loan bars inform credit underwriting decisions.

Filters: Sector, District, Gender.

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11. Key Insights

The following 12 decision-oriented findings are drawn directly from the four dashboards:

1. Portfolio Stress is Near-Critical

With 863 late and 399 default accounts, 47.8% of the portfolio is in repayment stress. This far exceeds healthy MSME NPA benchmarks and demands urgent remediation.

2. Tourism is the Riskiest Sector

At ~16% default rate, tourism-linked loans are the most vulnerable — likely reflecting post-pandemic demand disruption and the seasonal nature of J&K tourism revenue.

3. Agriculture Carries Structural Risk

A 15% default rate in Agriculture signals weather/market dependency. Crop insurance linkage and flexible repayment schedules are absent from the current design.

4. DLIC Approval Is the TAT Bottleneck

The DLIC approval stage alone accounts for disproportionate TAT across all districts. Streamlining this single stage could halve the average 188-day turnaround.

5. Doda Dominates Loan Volume

Doda leads in both total and average loan amounts, suggesting either higher project costs or stronger administrative capacity. Cross-district equitable allocation should be revisited.

6. EMI Ratio at 0.30 is Sustainable but Needs Monitoring

An average EMI ratio of 0.30 is within acceptable limits, but the income distribution peak at Rs. 50-70K/month means borrowers taking larger loans may be over-leveraged.

7. Loan Fully Funds Project Cost

Near-perfect correlation between loan amount and project cost indicates zero borrower equity contribution. This increases moral hazard and default incentive.

8. Female Rural Applicants Are a Majority Segment

The Gender x Residential breakdown shows female rural applicants are a dominant segment — a positive inclusion outcome, but one requiring tailored support given lower income bases.

9. Risk Scores Show Minimal Differentiation

Average risk scores range only from 47-57 across all gender-district cells, suggesting the risk model lacks discriminatory power to effectively identify pre-default signals.

10. Specially Abled Segment Is Underserved

1,109 of 1,501 specially-abled applicants have 'Unknown' disability type, indicating poor data collection at intake — a barrier to targeted disability inclusion programming.

11. Nano Entrepreneurs Dominate

Nano Entrepreneurs are the largest enterprise type in IT (121) and many other sectors, confirming the scheme is reaching micro-scale businesses as intended.

12. Pulwama Shows Highest Per-Stage TAT

Pulwama consistently records the longest stage-wise TAT, suggesting district administrative or banking capacity constraints requiring structural support.

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12. Recommendations

Based on the 12 insights above, the following 5 recommendations are proposed for immediate action:

R1. Introduce Sector-Differentiated Risk Underwriting

Apply stricter income-verification and project viability assessments for Tourism and Agriculture loans. Mandate crop/weather insurance linkage for agricultural borrowers.

Expected Impact: 3-5% reduction in sector default rates within 2 cohorts.

R2. Digitise and Fast-Track DLIC Approval

The DLIC approval stage is the single largest TAT driver. A digital approval portal with auto-escalation after 15 working days could reduce average TAT from 188 to below 90 days.

Expected Impact: 50%+ TAT reduction; improved borrower experience and fund utilisation.

R3. Mandate Minimum Borrower Equity Contribution

Require 10-15% owner equity for loans above Rs. 10 Lakh to reduce moral hazard. This aligns borrower incentives with repayment and reduces the scheme's credit exposure.

Expected Impact: Estimated 2-4% reduction in default rate among larger-ticket borrowers.

R4. Launch a Repayment-Risk Early Warning System

Use Risk Score, EMI Ratio, and sector flags to generate monthly watchlists of at-risk accounts. Assign field counsellors for early intervention 60+ days before due date.

Expected Impact: Convert ~15% of potential defaults to restructured/performing status.

R5. Improve Inclusion Data Quality at Application Stage

Mandate capture of Disability Type, exact Residential classification, and precise Gender at onboarding. This will enable scheme impact measurement and targeted benefit disbursement.

Expected Impact: Unlocks disaggregated policy interventions for 1,500+ specially-abled applicants.

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13. Limitations and Next Steps

Data Limitations

- No time dimension available — temporal trend analysis (e.g., seasonal default patterns) is not possible.
- Credit Score and Risk Score both average ~50 — possible dataset simulation artefact; external validation needed.
- Large 'Unknown' categories in Gender, Residential, and Disability fields limit sub-group analysis.
- No income verification flag — self-reported income may overstate borrower capacity.

Method Limitations

- No predictive model built — findings are descriptive; causality cannot be established from cross-sectional data.
- EMI Ratio assumes uniform 60-month tenure; actual repayment schedules may vary.
- TAT analysis relies on calendar-day counts without accounting for public holidays.

Suggested Future Work

- Build a logistic regression / gradient-boosted default prediction model using cleaned features.
- Integrate satellite/remote-sensing data for agricultural risk stratification by plot and season.
- Longitudinal cohort tracking: link application ID to actual repayment timelines from bank records.
- Geo-spatial mapping of loan concentration vs district-level economic indicators (GDDP, employment).
- Sentiment analysis on field officer notes (if available) to surface qualitative risk signals.

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14. Contribution Matrix

Team Member	Roll No.	Primary Responsibility	Key Deliverables
Sibtain Ahmed Qureshi, Anant Singh		Data Cleaning & KPI Design	Cleaned dataset, KPI definitions document
Sibtain Ahmed Qureshi, Bhawana Yadav		Dashboard 1 & 2 (Power BI)	DB-1 Loan Overview, DB-2 Repayment & Risk .pbix
Bhargav Patil, Sibtain Ahmed Qureshi		Dashboard 3 & 4 (Power BI)	DB-3 Applicant Profile, DB-4 Financial Deep-Dive .pbix
Hardik Hathwal, Kabir Sharma		Report & Presentation	project_report.docx, slide deck
Rohan		Statistical Analysis & QA	Analysis notebook, QA checklist

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